

Follicle-stimulating hormone (as its name implies) stimulates follicle growth, aided by LH, and the cells of the growing follicles start to make estradiol. Estradiol concentration slowly rises during most of the *follicular phase* (days 0–14), the part of the ovarian cycle during which follicles grow and oocytes mature. (Several follicles begin to grow with each cycle, but usually only one matures; the others disintegrate.) A low concentration of estradiol inhibits secretion of pituitary hormones, keeping the concentration of FSH and LH relatively low. In this portion of the cycle, regulation of the reproductive hormones closely parallels the regulation in males.

When estradiol secretion by the follicle begins to rise steeply, the levels of FSH and LH increase markedly. Why? Whereas a low level of estradiol inhibits secretion of pituitary gonadotropins, a high concentration has the opposite effect: It stimulates gonadotropin secretion by causing the hypothalamus to increase output of GnRH. A high estradiol concentration also increases the GnRH sensitivity of LH-releasing cells in the pituitary, further increasing the LH level.

The maturing follicle, containing a fluid-filled cavity, enlarges to form a bulge at the surface of the ovary. The follicular phase ends at ovulation (day 14), about a day after the LH surge. In response to FSH and the peak in LH level, the follicle and adjacent wall of the ovary rupture, releasing the secondary oocyte. At or near the time of ovulation, women may feel a pain in the lower abdomen, on the same side as the ovary that released the oocyte.

The *luteal phase* (days 15–28) of the ovarian cycle follows ovulation. LH stimulates the remaining follicular tissue to form the corpus luteum, a glandular structure. Stimulated by LH, the corpus luteum secretes progesterone and estradiol, which in combination exert negative feedback on the hypothalamus and pituitary. This feedback greatly reduces LH and FSH secretion, preventing maturation of another egg when a pregnancy may be under way.

If pregnancy does not occur, the low gonadotropin concentration at the end of the luteal phase cause the corpus luteum to disintegrate, triggering a sharp decline in estradiol and progesterone concentrations. This decline liberates the hypothalamus and pituitary from negative feedback. The pituitary can then secrete enough FSH to stimulate the growth of new follicles, initiating the next ovarian cycle.

The Uterine (Menstrual) Cycle

Prior to ovulation, ovarian steroid hormones stimulate the uterus to prepare for support of an embryo. Estradiol secreted in increasing amounts by growing follicles signals the endometrium to thicken. In this way, the follicular phase of the ovarian cycle is coordinated with the *proliferative phase* (days 6–14) of the uterine cycle. After ovulation, the estradiol and progesterone secreted by the corpus luteum stimulate maintenance and further development of the uterine lining, including enlargement of arteries and growth of endometrial glands. These glands secrete a nutrient fluid that can sustain

an early embryo even before it implants in the uterine lining. Thus, the luteal phase of the ovarian cycle is coordinated with the *secretory phase* (days 15–28) of the uterine cycle.

Once the corpus luteum has disintegrated, the rapid drop in ovarian hormone concentration causes arteries in the endometrium to constrict. Deprived of its circulation, the uterine lining largely disintegrates, releasing blood that is shed along with endometrial tissue and fluid. The result is menstruation—the *menstrual flow phase* (days 1–5) of the uterine cycle. During this phase, which usually lasts a few days, a new set of ovarian follicles begin to grow. By convention, the first day of flow is designated day 1 of the new uterine (and ovarian) cycle.

About 7% of women of reproductive age suffer from a disorder called **endometriosis**, in which some cells of the uterine lining migrate to an abdominal location that is abnormal, or **ectopic** (from the Greek *ektopos*, away from a place). Having migrated to a location such as an oviduct, ovary, or large intestine, the ectopic tissue responds to hormones in the bloodstream. Like the uterine endometrium, the ectopic tissue swells and breaks down during each ovarian cycle, resulting in pelvic pain and bleeding into the abdomen. Researchers have not yet determined why endometriosis occurs, but hormonal therapy or surgery can be used to lessen discomfort.

Menopause

After about 500 cycles, a woman undergoes **menopause**, the cessation of ovulation and menstruation. Menopause usually occurs between the ages of 46 and 54. During this interval, the ovaries lose their responsiveness to FSH and LH, resulting in a decline in estradiol production.

Menopause is an unusual phenomenon. In most other species, females and males can reproduce throughout life. Is there an evolutionary explanation for menopause? One intriguing hypothesis proposes that during early human evolution, undergoing menopause after bearing several children allowed a mother to provide better care for her children and grandchildren, thereby increasing the chances for survival of individuals who share much of her genetic makeup.

Menstrual Versus Estrous Cycles

In all female mammals, the endometrium thickens before ovulation, but only humans and some other primates have menstrual cycles. In other mammals, both domesticated and wild, the uterus reabsorbs the endometrium in the absence of a pregnancy, and no extensive fluid flow occurs. For these animals, the cyclic changes in the uterus occur as part of an **estrous cycle** that also controls the sexual receptivity of females: Whereas human females may engage in sexual activity throughout the menstrual cycle, mammals with estrous cycles usually copulate only during the period surrounding ovulation. This period, called estrus (from the Latin *oestrus*, frenzy, passion), is the only time the female is receptive to